Michael Fischbach

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Top Secret Clearance (In Process) | SCI Eligible | Malibu, CA

OBJECTIVE

Mechanical Engineering student (Cal Poly Honors, 3.79 GPA) seeking a mechanical design or manufacturing internship. Hands-on CAD-to-part experience across 3-axis CNC machining, prototyping, tolerance analysis, and iterative real-world testing. Ran an independent manufacturing business from 2019 to 2026, so the design-workshop-testing loop is something owned end to end rather than observed.

EDUCATION

California Polytechnic State University, San Luis Obispo Sep 2025 – Expected May 2028

B.S. Mechanical Engineering, Honors Program | GPA: 3.79 | Cal Poly Golf Engineers & Agentic Engineering

Coursework: Subtractive Manufacturing (IME 145/146), Materials Joining, Materials Engineering with lab; in progress (Fall 2026): Statics, Design Using Solid Modeling (SolidWorks), Applied Programming for ME

- Honors research-track program; Learn by Doing curriculum with hands-on manufacturing, CNC machining, and design from first year
- Built multi-component mechanisms (piston, Geneva wheel, CNC-machined screwdriver) with tolerance stack-up analysis and GD&T
- Designed a low-cost, accessible door handle for one-handed users in Cal Poly's Red Brick dorms; PLA base, TPU head

Oaks Christian School – Institute of Engineering Sep 2022 – Jun 2025

Engineering Pathway: 6 multi-month applied design projects | GPA: 3.92 UW / 4.10 W *Malibu, CA*

EXPERIENCE

WERC Design Contest Team Lead – NASA Artemis Wireless Capacitive Charger 2026 – Present

Cal Poly SLO *San Luis Obispo, CA*

- Lead a 12-person graduate/undergraduate team developing a 1 kW capacitive charger for NASA's Artemis lunar rover
- Set schedule and integrate power, coupler, controls, and structures; develop plate-based alignment sensing for rover docking
- Targets: 80%+ efficiency across a 100 mm gap at full scale, 25 W at 70%+ on the bench; dust, vacuum, -203°C to 54°C

Mechanical Engineering Intern 2025 – Present

All Domain Integration (ADI) – Aerospace Engineering *Remote*

- **SCI eligible · Submitted for Top Secret clearance**
- Support mission systems engineers on Space Force and Air Force efforts as part of a SETA contractor team
- Use locally hosted large language models (LLMs) to analyze satellite sensor data and draw knowledge-based conclusions

Founder & CEO 2019 – 2026

Chameleon Ramps · chameleonramps.com *Malibu, CA*

- Designed 40+ concrete fingerboard obstacles ($10–$50 each); proprietary molding process cut material waste ~40%
- Generated $39,600 in revenue across 627 orders by running a design-test-refine loop of 4–6 prototype generations per product, modeling in Fusion 360 and 3D printing prototypes to build robust, reusable molds for repeat production
- Grew revenue 30% via a 7-creator sponsorship program; co-designed Recycled Ramps collaboration; shipped to 15 countries
- Cast concrete range produced entirely in-house, mold design included; the quarter pipe was the hardest mold to release, which drove a redesign of the draft angles

CEO & Founder Jun 2026 – Present

Campus Native · campusnative.com *San Luis Obispo, CA*

- Founded and ran a digital marketplace connecting students with campus tour guides
- Recruited a 4-person intern team across engineering, marketing, and content; set their priorities, reviewed their work, led the build of the listing and booking flow

ENGINEERING PROJECTS

CNC-Milled Golf Putter – Full CAD-to-Part Build 2025

3-axis Haas CNC, Fusion 360, DFM · Oaks Christian capstone

- Designed and machined a novel putter from 6061 in two ops with adaptive toolpaths validated with 12 PLA prototypes
- Designed a scooping cavity whose undercuts drove the toolpath strategy; cut the 3-degree loft angle on a 3D-printed shim
- Modeled unit cost at $18.56 vs. $250 retail; 1 of 8 finalists (40+ students), presented to 200+ alumni at Capstone Night

Wave Energy Converter – Team Lead 2023

Bridgeport mill, Fusion 360, fluid dynamics · Oaks Christian

- Led a five-person team to a 1st place win of five teams, designing and machining a 1:12-scale rack-and-pinion wave energy converter, with all parts made to tolerance in-house
- Prototype produced 120 V at 1:12 scale (15 V per motor), projecting 360 V at full scale
- Applied fluid dynamics and energy-transfer analysis; modeled the full assembly in Fusion 360 and AutoCAD

Apricot Lane Farms – Net-Zero Carbon Shipping Container Space Mar – May 2024

AutoCAD, Autodesk CFD, Project Management · Oaks Christian

- Selected as 1 of 3 students to design a net-zero carbon workspace for Apricot Lane Farms; farm chose the L-shaped design
- Analyzed structural integrity of container walls with cutouts and roof overhang; assessed solar needs with the farmers
- Drafted the floor plan in AutoCAD with an ADA-accessible restroom and incinerating toilet; built a 1:10-scale model

Water Distribution System – Cerro de Agua, Nicaragua Mar – Jun 2023

Autodesk CFD, Geospatial Analysis, Cost Analysis · Oaks Christian

- Designed a gravity-fed water distribution network for a village in Nicaragua, including tank siting from a site survey
- Selected 1 of 5 main presenters from 40 classmates; presented to Cerro de Agua city officials, school administration, and engineering alumni

TECHNICAL SKILLS

CAD / CAE: Fusion 360 (6+ years, 55+ designs), SolidWorks (2+ years), AutoCAD, Autodesk CFD

Manufacturing: 3-axis Haas CNC machining, Bridgeport mill, lathe, FDM/SLA 3D printing, composite & concrete casting

Methods: GD&T (ASME Y14.5), DFM, tolerance stack-up analysis, iterative prototyping, design-test-refine, cost modeling

AI: Claude Code, Codex, and Gemini via agentic workflows in Antigravity IDE

Software: Python, Git, MATLAB, Linux, HTML/CSS

Certification: Haas CNC Certification Program – Basic Mill Operator, online coursework completed (Haas Automation, 2026)

Languages: English (native); German (beginner, actively learning)

AWARDS

- **Capstone Finalist (CNC putter)** – 1 of 8 from 40+ students, Oaks Christian, 2025
- **1st Place, Wave Energy Converter** – 1st of 5 teams, Oaks Christian, 2023
- **President's Honors List** – Cal Poly SLO, 2026; awarded for Dean's List three consecutive quarters
- **Cal Poly Honors Program** – selective research-track admission, 2026
- **1st Place, Noodle Bridge Design** – Cal Poly SLO, Nov 2025; won by a 38.5% margin

LEADERSHIP & ACTIVITIES

- **Eagle Scout** – Boy Scouts of America (2025); Senior Patrol Leader; 15 years in the program
- **Internationally Mobile** – U.S. Air Force family; 9 relocations across 7 states and territories
- **Volunteering** – Meal Outreach, St. Aidan's Church (5+ years)

INTERESTS

Backpacking – 500 mi | Weightlifting – 355 lb squat | Poker – 3rd of 126, Cal Poly tournament | Watercolor/Drawing – 3 contest wins